Predicting Housing Prices Case Study Hook Document

Welcome to the Charlottesville Housing Analytics Lab. You have been brought on as a junior data analyst to help address a growing concern in the local housing market. City officials and policy advisors are struggling to anticipate changes in housing prices, and traditional forecasting models based only on past prices are often too slow to capture emerging trends. At the same time, the Housing Advisory Committee meets regularly to discuss issues like affordability, zoning, and development. These conversations often reflect concerns, uncertainty, or optimism about the market – but no one knows whether they can actually be used to predict what happens next.

Your mission: investigate a real dataset of housing market reports and Housing Advisory Committee meeting minutes to determine whether the language used in these discussions can improve forecasts of housing prices. Your job is to build a forecasting pipeline that combines structured numerical data with unstructured text data, transforming meeting minutes into meaningful features and evaluating whether they add predictive value beyond traditional models. This task centers on extracting sentiment from text, aligning it with time-series data, and testing whether it improves out-of-sample predictions.

Through this case study, you will explore how text preprocessing and simple sentiment measures can be integrated into time-series models like ARIMA and ARIMAX. You will analyze how different modeling choices impact forecasting performance, examine where models succeed and fail, and decide whether sentiment from policy discussions provides a meaningful signal or just noise. Can the tone of local policy conversations reveal where housing prices are headed? City officials are waiting for your insight.

Your toolkit includes housing market data, meeting minutes, Python scripts, supplemental articles, and the project rubric. To begin, clone the repository and start by reading the README, which outlines the workflow, file structure, and the steps needed to reproduce the analysis.

Everything you will need can be found here:
https://github.com/emilyjmoore/DS_4002_CS3